Generative AI for End-to-End Limit Order Book Modelling

A Token-Level Autoregressive Generative Model of Message Flow Using a Deep State Space Network

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Why do we Need Generative Microstructure Models?

Actions in limit order books (LOBs) have consequences: **market impact**

↳ generative black-box solution to adaptive order flow

Microstructure Messages as Language

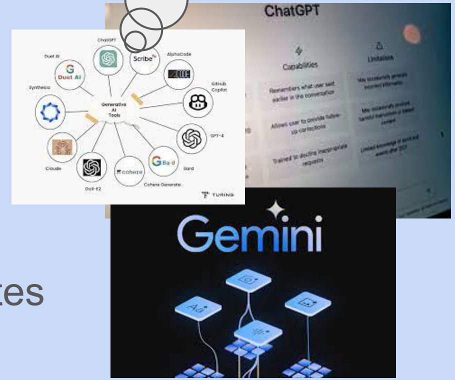
Model data on most **granular** scale

↳ Difficult with CGANs…

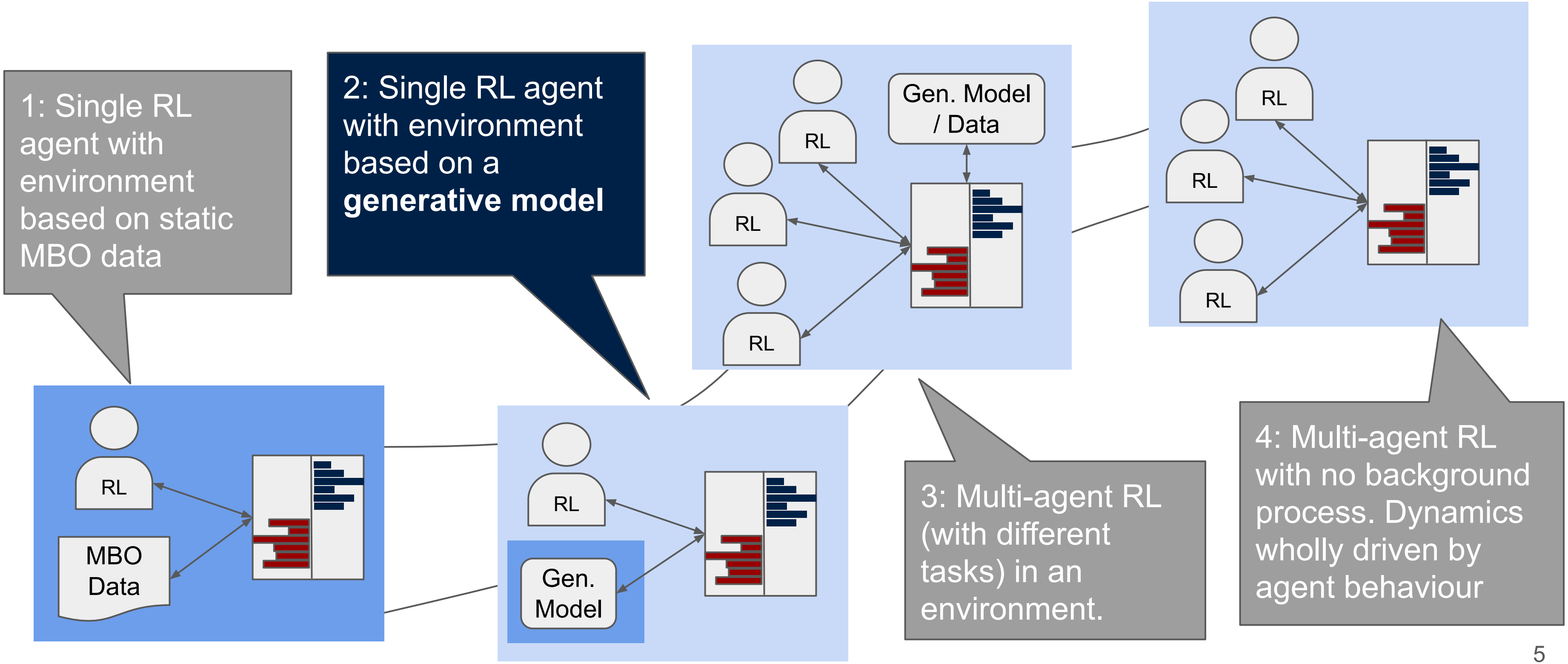
↳ **Autoregressive** generative model for LOB **messages**

  ↳ Similar to LLMs

  ↳ trained on tokenized message sequences and LOB states



The Road to Fully RL-Based Agent Populations



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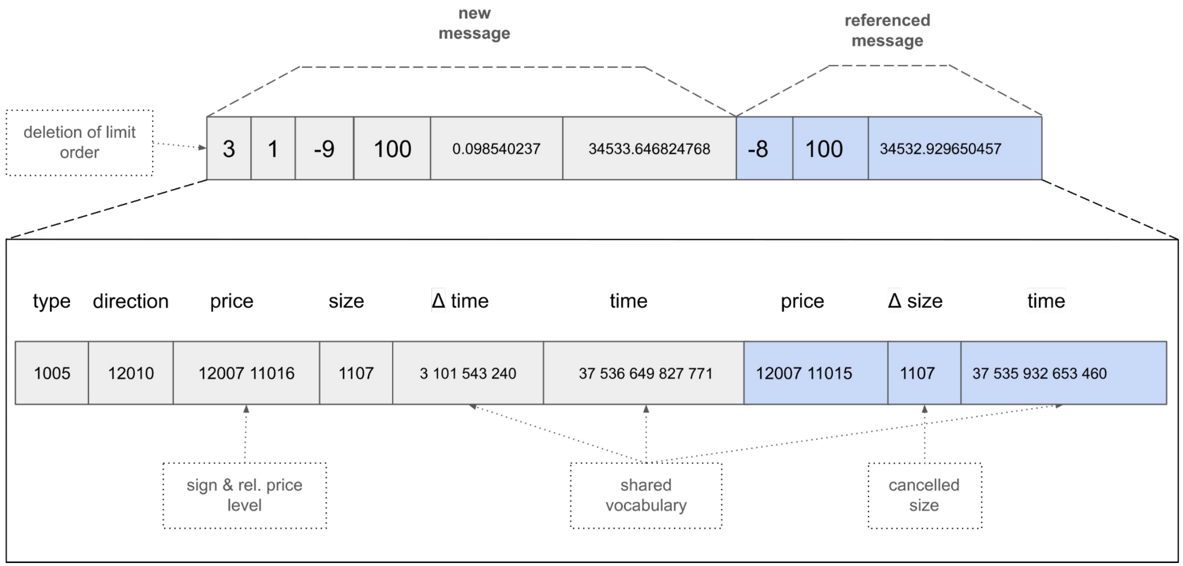
Message Data is Tokenized



Using NASDAQ LOB data (**LOBSTER**), we develop a **custom tokenizer for message data**, converting groups of successive digits to tokens, similar to tokenization in large language models.

S5 Layer Architecture

Discretization of continuous linear **State-space model**:

$x'(t) = Ax(t) + Bu(t)$

$y(t) = Cx(t) + Du(t)$

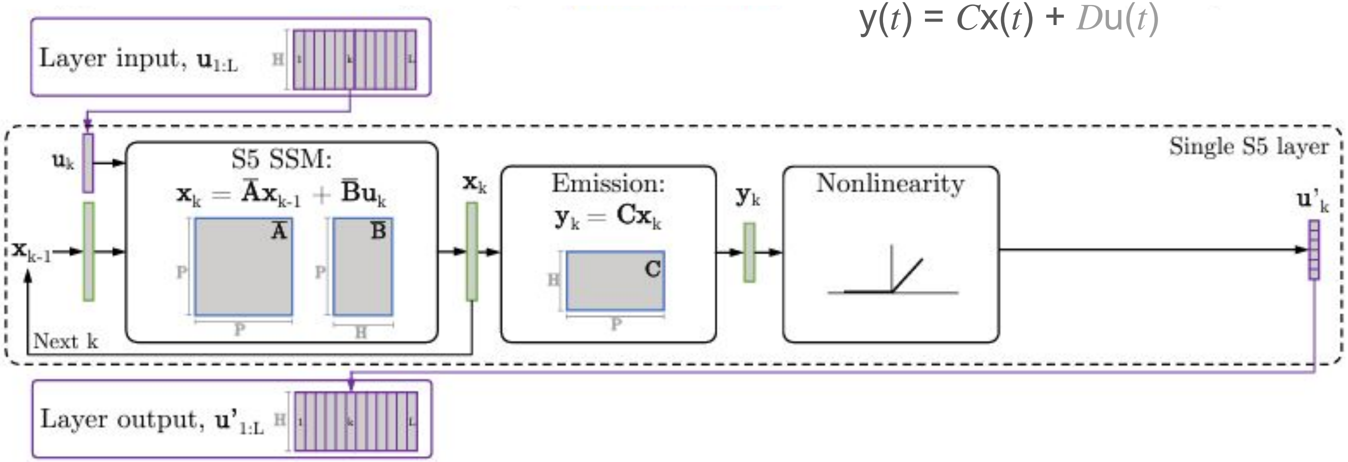
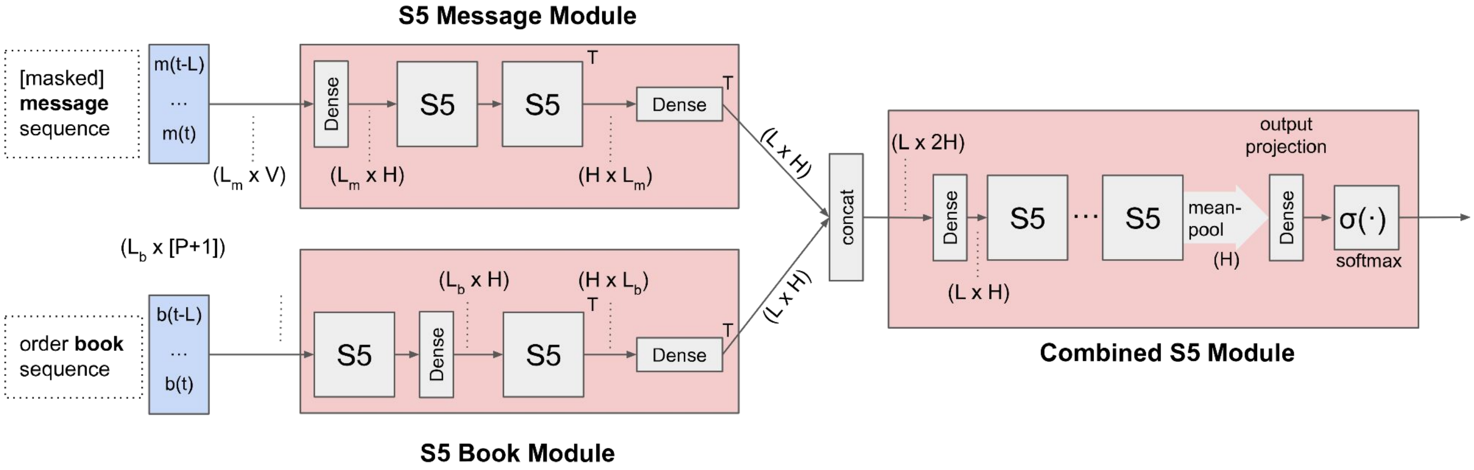


Figure 2 from Jimmy TH Smith, Andrew Warrington, and Scott Linderman. 2022. Simplified State Space Layers for Sequence Modeling. In The Eleventh International Conference on Learning Representations.

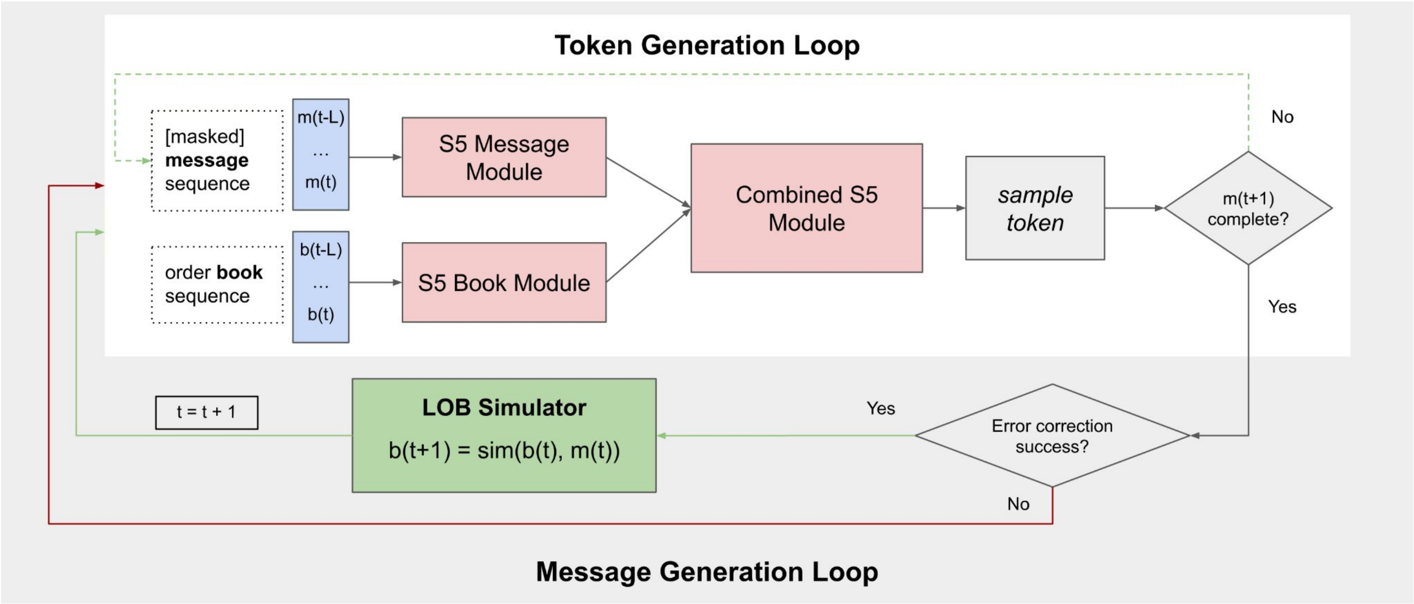
To manage long sequences, the model employs **simplified structured state-space layers (S5)** to process **tokenized messages** and **sequences of order book states**.

Message and Book Data is Combined



To manage long sequences, the model employs **simplified structured state-space layers (S5)** to process **tokenized messages** and **sequences of order book states**.

JAX-LOB Interprets Generated Messages



By embedding the model in an inference loop with JAX-LOB as a market replay simulator, we are able to generate **full-fidelity granular trajectories of entire LOBs**.

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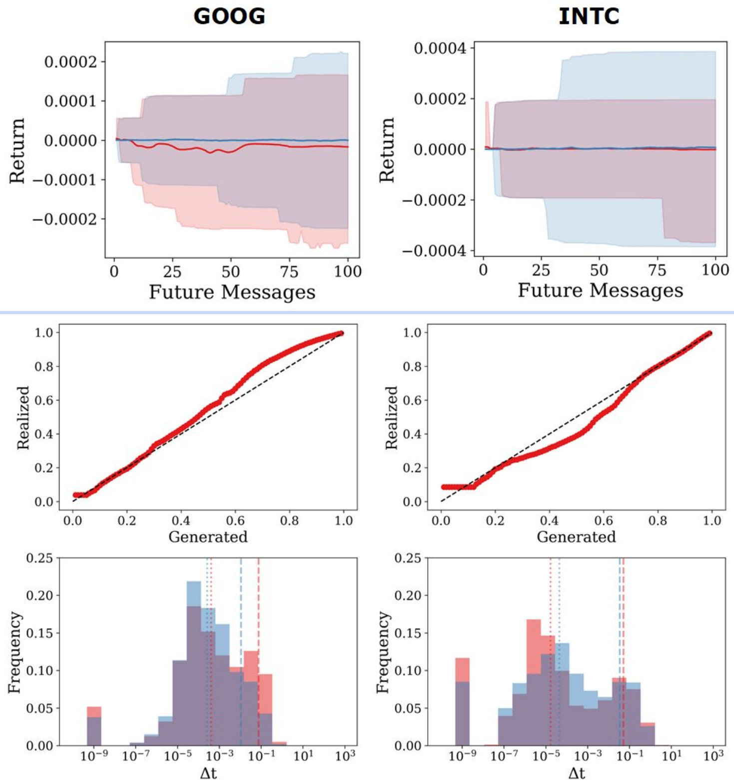
Matching Empirical Distributions



generated
realized

Future mid-price return distributions

QQ plot

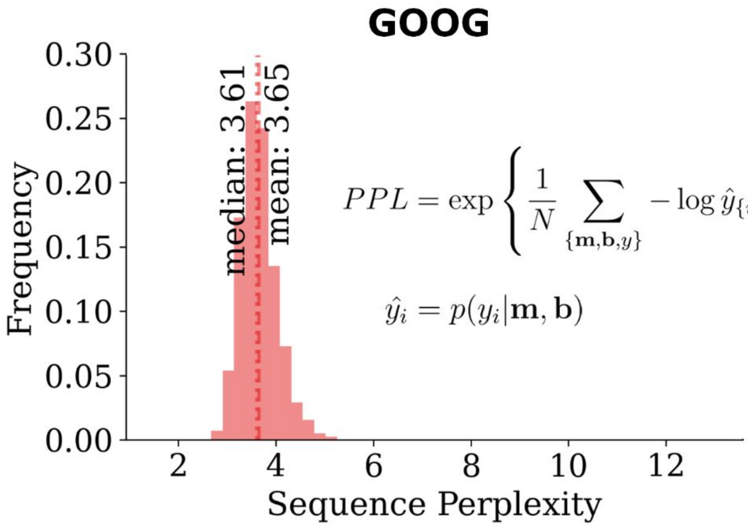
Histogram

Message inter-arrival times

The model matches aggregate data distributions by training on micro-structure tokens.

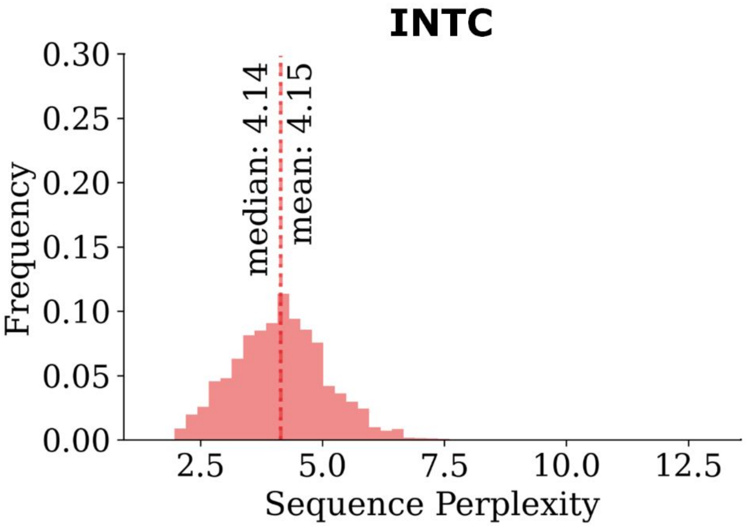
Times are treated the same as other features.

Low Sequence Perplexity (PPL) Indicates Good Fit



$$PPL = \exp\left\{\frac{1}{N}\sum_{\{\mathbf{m},\mathbf{b},y\}} -\log \hat{y}_{\{i=y\}}\right\}$$

$$\hat{y}_i = p(y_i|\mathbf{m}, \mathbf{b})$$



Generated 100-message sequences exhibit **low out-of-sample model perplexity**.

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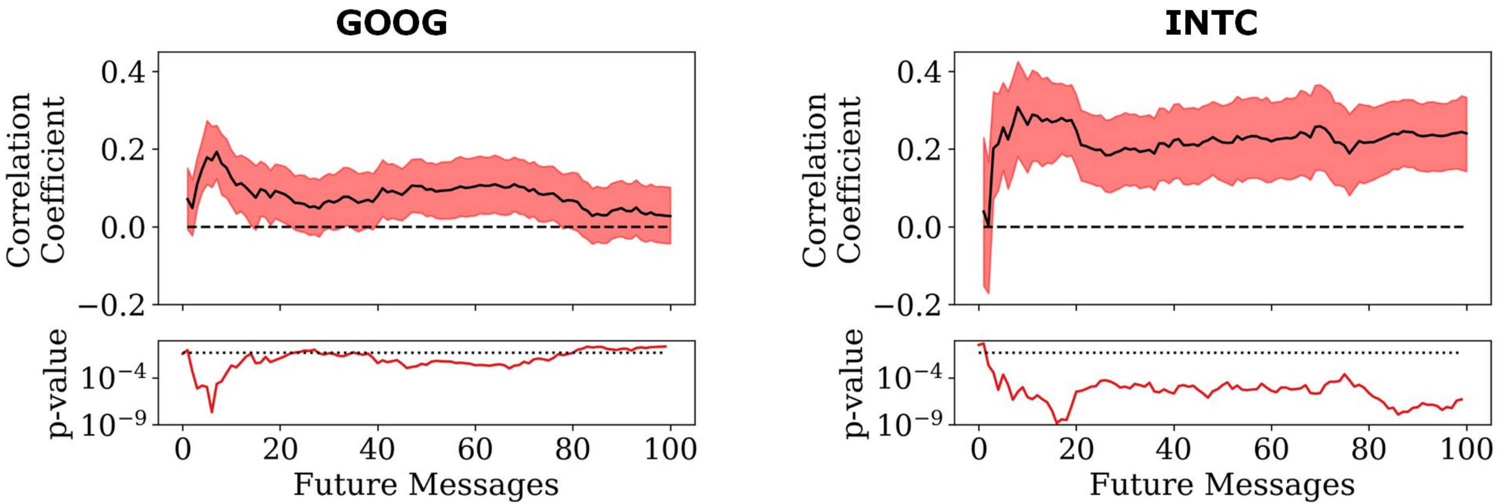
Generated Forecasts Predict Future Returns



The mid-price **returns** calculated from the generated order flow exhibit a **significant correlation** with the data, indicating impressive **conditional forecast performance**.

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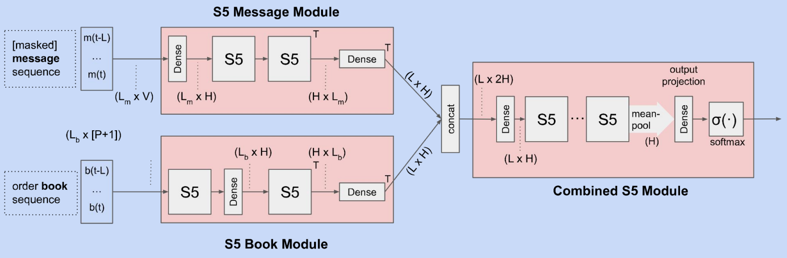
Open Issues and Future Work

- **Semantic errors** in generated messages
  e.g. reference to non-existent prior order
- **Long sequence** generation

- **Larger model** size to address these?
- Larger **empirical validation** of microstructure characteristics (impact, correlations etc.)



Currently using **6 S5 layers**, resulting in only ~ **6.3x10^6 parameters**

Conclusions

- **Custom data tokenization** allows analogous training to large language models.
- Our **autoregressive** paradigm generates granular microstructure messages, instead of aggregate returns.
- **Granularity** offers new application areas: world models or generative environments in high-frequency financial RL.

Autoregressive Large Financial Environment Models

Questions?

https://dl.acm.org/doi/pdf/10.1145/3604237.3626898

https://github.com/peernagy/LOBS5

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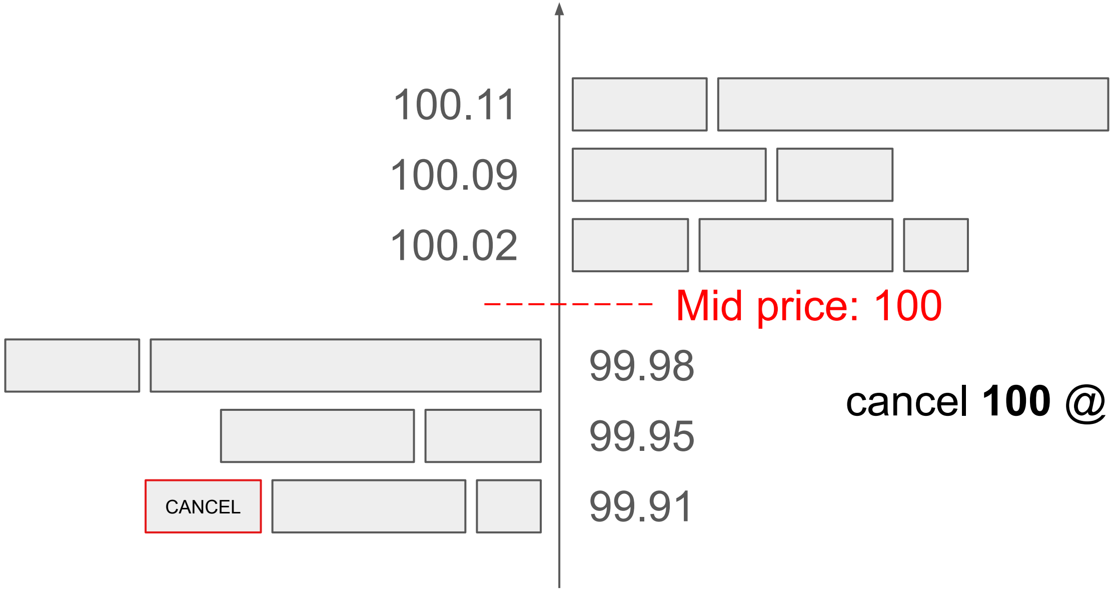
Market by Order Data : **LOBSTER**



cancel **100** @ price $99.91 from order *8574201*

cancel **100** @ price -9
from order submitted at price -8 at time t_{ref}